UNIVERSITY OF LIMERICK
OLLSCOIL LUIMNIGH

Master of Science in Artificial Intelligence

Project and Dissertation Guidelines

Academic Year 2025/26

TABLE OF CONTENTS

1 Introduction 1
2 Project Calendar 2
3 Project Identification 3
4 Major Milestones 5
5 Intellectual Property 7
6 Confidentiality 7
7 Project Extension Requests 8
Appendix A Autumn Year 2: Project Report Details 9
Appendix B Autumn Year 2: Supervisor Meetings 10
Appendix C Spring Year 2: Progress Project Report 10
Appendix D Summer Year 2: Dissertation 11
Appendix E Summer Year 2: Final Project Presentation 14
Appendix F Grading 14
Appendix G Major Update 15

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1 INTRODUCTION

In this document, the requirements for the MSc in AI project and dissertation are outlined. The word dissertation implies an extended piece of writing on the results of an independent investigation of a question or proposition. The MSc project is undertaken throughout the full calendar year and it is expected that progress will be made in the Autumn and Spring semesters, and the project will be finished in the Summer semester (May-August). The objective of the project is to give a student the chance to research and develop a topic in depth and to apply his/her theoretical and practical knowledge to a practical problem.

Students will start to think about project areas and ideas in the module ***CS6163*** *Advanced Topics Seminars and Project Specification.* Supervisors in parallel will identify areas of interest and projects.

With a topic identified, students will undertake a literature review related to the project area, and supervisors will be allocated.

The main project work will be undertaken throughout the second year resulting in the final presentation, and project grading within the module *ED5005 Project and Dissertation - Artificial Intelligence.*

Note: The Industry Advisory Board for the MSc in AI has advised that the code resulting from dissertation projects can be an effective way of showcasing your knowledge and skills in support of applications for new professional roles. A Github repository is an excellent way of creating such a showcase.

2 PROJECT CALENDAR

The calendar below identifies the two modules associated with the project and allocates the project related tasks and milestones across them.

Table 1: Project related deadlines

MSc in Artificial Intelligence: Project Modules			Period	Start	Finish
1	CS6163	Advanced Topic Seminars and Project Identification	Spring/Summer Y1	26-Jan-26	June 2026
2	ED5005	Project and Dissertation - Artificial Intelligence	All Yr2	Sept-26	August 2027

Project Tasks and Major Milestones AY 2025/26			Period	Start	Finish
1	CS 6163	Advanced Topics Seminars and Project Identification	Spring/Summer Y1	27-Jan-26	June 2026

		Project Tasks and Major Milestones AY 2025/2026	Semester	Week/Date	Date
1	CS 6163	Supervisor and Project List made Available	Spring Yr 1	1-5	Jan-26
2	CS 6163	Supervisor and expert presentations		March 31	various
3	CS 6163	Assignments 1-4 - See module page for individual deadlines	Spring/Summer Yr 1	1-5	May/June 2026
4	CS 6163	Confirm Project/Topic	Summer Yr 1	1	31-May-26
		Intellectual Property: Please note that UL cannot engage in Non-Disclosure agreements to protect company IP shared as part of the masters projects			
5	CS 6163	Supervisor: Allocation			May/June

Project Tasks and Major Milestones AY 2026/27			Semester	Week	Date
1	ED5005	Project Report: Finalised Literature Review, Research Question	Autumn Yr 2	0	6-Sep-26
2	ED5005	Report Supervisor Meeting: Deliverables, roadmap, Gantt related to resolving project dependencies (e.g data collection)		2	20-Sep-26
3	ED5005	Report Supervisor Meeting: Agreed project tasks and Gantt chart		14	13-Dec-26
4	ED5005	Project Report Update: Option 1	Spring Yr2	1	29-Jan-27
5	ED5005	Project Report Update: Option 2		15	16-May-27
6	ED5005	Dissertation Submission Option 1	Summer Yr 2		31-May-27
7	ED5005	Final Presentation Option 1		7-Jun-27 to 11-Jun-27	
8	ED5005	Dissertation Submission Option 2			1-Aug-27
9	ED5005	Final Presentation Option 2		16-Aug-27 to 20-Aug-27	

3 PROJECT IDENTIFICATION

Students are expected to own and drive their project. Each is responsible for developing and managing the project from determining the main objectives, to scoping and planning, design, implementation and evaluation. There is huge diversity of approaches on how projects are identified. Some students will already have a project from prior experience; some will have narrowed to a specific area; others will have several ideas and many at this stage will have no idea of even how to get started. Hence, an iterative process of student and supervisor project assignment will be adopted.

3.1 STUDENT

Students are strongly encouraged to propose their own projects (locally known as homebrews). In addition, it is anticipated that many projects will be employment or externally generated and supported. As part of CS6163, students will be asked to outline the project or to identify potential areas of interest.

If a student plans to undertake an already known homebrew/employer/industry centred project, they must give outline details including a title; the AI area(s) addressed; a brief description; identify any major resources required e.g. datasets, high-end computing, toolsets, etc and where these are available; the company mentor and first reader where appropriate.

3.2 SUPERVISOR

The academic supervisor's role is to provide guidance to the student during the project. Supervisors are there to help with ideas and to give advice, they are not there to teach fundamentals nor to time manage. They will be primarily responsible for grading the project as part of a project panel.

You will be assigned a supervisor at the start of the project and any request for change in supervision needs to be brought to the programme team.

3.3 EMPLOYER/INDUSTRY

Where an employment or industry based project is identified, the role of the company mentor is to help identify and specify a suitable project; support the student with the implementation (timely response to email queries from students, guidance in dataset preparation, help with a statement of work, assessing intermediate project outputs); provide a first reader for reports and dissertation; ensure support of others; and satisfy themselves in relation to confidentiality. An embargo on access to the dissertation can be placed to ensure confidentiality. It is the student's responsibility to ensure the availability of the company mentor and to ensure an embargo is in place if so required by the company. Embargos can be requested when submitting the final dissertation.

3.4 PROJECT TOPIC CHANGE

Changing your dissertation topic can be a significant decision, and it's essential to approach it thoughtfully.

Should you decide to change your project topic, you must discuss this initially with your supervisor, explain the reasons for the change and seek their feedback. Take into consideration any valuable insights or concerns raised by your supervisor.

Depending on the change, reassignment of supervisors may be necessary for the expertise in a new project area. For a supervisor change, the change of project must be communicated to the programme team with a request for a new supervisor. The final decision re supervisor change lies with the programme team and is subject to the availability of supervisors.

4 MAJOR MILESTONES

Throughout Year 2, a number of major project milestones and deliverables are defined. The associated deadlines can be found in Table 1.

1. At the beginning of Year 2, a project report will be completed and supervisor meeting will take place resulting in Supervisor Meeting Report 1.
2. At the end of the Year 2 autumn semester, a further supervisor meeting will take place resulting in Supervisor Meeting Report 2.
3. In the Year 2 spring semester, a project status report is required, focusing on the progress to date, issues arising and their planned resolution.
4. At the end of Year 2, in the summer semester, the dissertation and final presentation will be completed. Two options are available for submission: Option 1 - early summer and Option 2 - late summer.

Table 2: Thesis stages and presentations

Autumn	**Project Report: Objectives, Literature Review, Plan** **Supervisor Meeting and Update 1** **Supervisor Meeting and Update 2**
Spring	**Project Report: Update and Progress Made**
Summer	**Dissertation Submission** **Project Presentation: Final**

4.1 AUTUMN YEAR 2: PROJECT REPORT -LITERATURE REVIEW AND PLAN

By this milestone, the student is expected to have a clear understanding of the project objectives, become familiar with the theoretical aspects of the project, and study the background literature. This will take the form of a review of related research, relevant projects, and include published articles, text books, previous project reports or theses. The supervisor will be able to guide the student on what is required.

The literature review and plan is due at the beginning of Year 2. Note that students will undertake a short literature review as part of the Research Methods module, where you will learn approaches, report structuring and referencing. It is expected that students will perform this literature review on their chosen dissertation subject.

The Autumn report will introduce and cover the objectives of the project, present work to date including the literature review, and give a plan of work for the remainder (with estimated completion dates for each major task), against which the student's performance can be assessed. All major risks should be identified, assessed and documented. More details are given in Appendix A.

4.2 AUTUMN YEAR 2: PROJECT SUPERVISOR MEETINGS

The work undertaken to date, and documented in the above report will be presented at two formal supervisor meetings e.g. in the form of a 10 minute presentation. Details are given in Appendix B.

4.3 SPRING YEAR 2: PROJECT REPORT UPDATE -PROGRESS

The Spring report will cover the progress made to date and present a revised plan (you should discuss substantial changes with your supervisor). It largely should be an update of the Autumn report. Previous feedback given by the supervisor should be addressed. Further details are given in Appendix C.

As show in Table 1, two options are available. Please note the implications of the two options under 4.6.

4.4 SUMMER YEAR 2: DISSERTATION

This is a comprehensive report which documents the project as detailed in Appendix D.

4.5 SUMMER YEAR 2: PROJECT PRESENTATION

The final presentation and demonstration will be an online 15 minutes session to an examination panel focusing on the objectives achieved. Further details are presented in Appendix E.

4.6 FINAL SUBMISSION OPTIONS AND DEADLINES

There are two options for completion of the thesis and associated presentation:

Option 1: Submission and presentation at the start of the summer.

Option 2: Submission and presentation at the end of the summer.

Please see Table 1 for information on the deadlines for both options. Students should agree the option to be pursued with their supervisors in advance and this decision should be reviewed in light of progress made. In general, students making use of Project Report Update Option 1 may choose Option 1 or 2 for the Dissertation Submission. Students making use of Project Report Update Option 2 will be expected to choose Dissertation Submission Option 2. Deviations from these guidelines should be discussed and agreed with your supervisor.

5 INTELLECTUAL PROPERTY

Intellectual Property is governed by the University of Limerick Intellectual Property Policy approved by the UL University Research Committee on 30 September 2020_ (policy link). This policy explains how UL may assert intellectual rights over UL research:

> *'UL Research' means activities that involves one or more* of *the following:*
>
> - *any use* of *Intellectual Property owned or controlled by UL;*
> - *any use* of *UL research resources, research facilities or research equipment, the use* of *which informs the development* of *Intellectual Property.*
> - *any use* of *UL research funding (internal or external) that is administered or hosted by UL;*
> - *any research commissioned by third parties under which UL is contractually obliged to ensure that it owns any and all IP arising from such activities.*

If the research is undertaken by the student on the company's premises then UL Policy is that UL does not make any claim on the IP. If the research undertaken by the student involves significant *use* of *UL research resources, research facilities or research equipment, the use of which informs the development of Intellectual Property,* then the UL Policy states that IP created falls under the definition on UL IP.

6 CONFIDENTIALITY

An embargo on access to the dissertation can be granted where material is confidential, similar to the research thesis embargo process. Normally this does not exceed 5 years. Initially, you should consult with your academic supervisor on the confidentiality requirement, and how to address it. If material presented in the dissertation is deemed to be confidential, a request for an embargo should be made to the course director. When normally granted (in some cases after discussion it is not necessary), the candidate should include the embargo information in the declaration section at the beginning of the dissertation. At the end of the embargo period, the dissertation may be made fully accessible unless the student requests to renew the embargo, which will be subject to review.

Please note that UL cannot engage in Non-Disclosure Agreements to protect company IP shared as part of the masters projects.

7 PROJECT EXTENSION REQUESTS

Due to the strict UL grading deadlines, project extensions can only be facilitated in exceptional circumstances. Please note that any extension will likely result in a delay of a year in graduation and will normally incur an additional fee.

Any request for an extension must follow the below process:

- A formal written application must be made to their supervisor and course director in advance of the submission deadline.
- Discuss and agree an extension with their supervisor in advance of the submission deadline.
- The agreed deadlines for dissertation submission and presentation should be agreed and submitted to the course director. Supervisors are responsible for arranging a presentation slot and examiner panel.

I-Grades

In case of immediate crisis situation (illness, family bereavement etc.), students may apply for an Igrade which will allow them to submit the following summer without incurring additional fees. (See section "4.2.3 I Grades" in the Handbook of Academic Regulations and Procedures).

If applying for an I-grade, please familiarise yourself with the procedures and associated deadline as early as possible. Visit the I-grades webpage.

APPENDIX A AUTUMN YEAR 2: PROJECT REPORT DETAILS

The interim Autumn and Spring reports should be typed, font size 12 Times New Roman, 1.5 line spacing. The report length is typically between 15 and 20 pages, but the supervisor can advise of exact requirements.

Sections of the Report:

1. **Front title sheet**
   Include the following information:
   Name, **I.D.** number, Supervisor's Name, Course Followed, Year, Department, Title of Project.
2. **Table of Contents**
3. **Introduction and report outline**
   Clearly documents the project aims and objectives. Includes a brief outline of the report.
4. **Literature survey**
   Review of the literature and the background to the work. Identify similar projects.
5. **Theory**
   Present the initial analytical and technical theoretical aspects of the work, if appropriate.
6. **High level design**
   If available, include some high-level designs, describing computing, software or process related blocks, as appropriate for the project.
7. **Detailed action plan**
   This can be in the form of a Gantt chart for the project, in particular showing all the major tasks and their estimated start date, end date and duration in days associated with the resolution of project dependencies (data collection, access to facilities, expertise,..).
8. **Discussion**
   A short section discussing the progress to date, any arising challenges and identified risks.
9. Requirements of facilities and materials.
10. References and sources of information.

Submitting the Autumn report:

Submit the electronic copy of the final report in pdf format to your Brightspace project module MSc in AI Module - ED5005 named as follows, where XY are the initials of your supervisor and xxxxxxxx your id number:

Autumn_**XY**_**xxxxxxxx**_FirstName_Surname.pdf

APPENDIX B AUTUMN YEAR 2: SUPERVISOR MEETINGS

Submission dates are hard and can be found in Table 1.

Notwithstanding our expectation that students will meet with their supervisor(s) more frequently, two supervisor meetings will be associated with a supervisor meeting report.

Report 1 -Autumn Yr 2 Week 2:

Students should present their work to-date to their supervisor(s) in a 10 minute presentation with an emphasis on deliverables and project roadmap including a Gantt related to resolving project dependencies (e.g. data collection). The report should cover this high-level plan and any agreed changes.

Report 2 -Autumn Y2 Week 14:

Students should present their work to-date to their supervisor(s) in a 10 minute presentation with an emphasis on detailed project tasks including a Gantt chart. This project plan should include a significant body of work planned for execution in between the Y2 Autumn and Spring semester.

In the presentations the student may use up to 6 slides and it should take no longer than 10 minutes to allow for questions. The quality of the presentation should be of a professional standard.

APPENDIX C SPRING YEAR 2: PROGRESS PROJECT REPORT

The report format **will** be similar to the Autumn report but clearly explaining progress made.

Submitting the Spring report

Submission dates are hard and can be found in Table 1. Please note the implications of choosing submission option 2 (Section 4.3). Submit the electronic copy of the final report in pdf format to your Brightspace project module MSc in AI Module - ED5005 named as follows, where XY are the initials of your supervisor and xxxxxxxx your id number:

Spring_XY_xxxxxxxx_FirstName_Surname.pdf

APPENDIX D SUMMER YEAR 2: DISSERTATION

An important part of the undertaken project is writing the dissertation. The dissertation must document clearly what the student has achieved by unambiguously indicating the student's contribution and information gained from other sources, using appropriate referencing. It may include material from the reports submitted in the Autumn/Spring semesters.

D.1 SECTIONS

The following section outline will normally apply.

1. Front title sheet
   Include the following information:
   Name, I.D. number, Supervisor's Name, Course Followed, Year, Department, Title of Project.
2. An optional title page
   Picture/Graphic representing project, Title of Project, Name, I.D. number, year.
3. **A single page Abstract**
4. **The Table of Contents**
   Follow with a List of Figures, List of Tables if appropriate.
5. **Introduction and Outline**
   Introduce the topic and context. Clearly state the objectives of the project. Include an outline of the rest of report: e.g. a few lines describing each Chapter.
6. **A number of core chapters which might include:**
   Literature Review (of Similar Work)
   Analytical Background/Requirements
   Analysis
   Specification and Design, Implementation, Testing,
   Results including measurements and facts about performance Discussion of results
7. **Conclusion**
   Summarise how the objectives were met. Recommendations for future work.
8. **References**
9. **Acknowledgements**
   You should thank those who supported the project and you.
10. **Appendices**
    Discuss the inclusion of appendices e.g. code listings etc. with your supervisor.

D.2 ALTERNATIVE FORMATS

In addition to the standard thesis as described in section D.1, the following formats may be chosen subject to explicit approval from your supervisor.

D.2.1 Annotated Jupyter notebooks

Instead of item 6 in section D.1 (Core chapters) you may include annotated Jupyter notebooks that fully document your project work. These notebooks should contain the main code contributions of your project with appropriate visualisations and should use Markdown cells (and its formatting capabilities) to provide a narrative to your code. The reviewer should be able to understand the key steps taken in your project, and your results, by reading the narratives alone.

D.2.2 Scientific Paper

You may structure your report as a 10-page double column IEEE style research paper. Such a paper will include all aspects from items 5-9 in section D.1 in a shortened fashion. The paper should be deemed to be of a sufficient standard to be submitted to a conference or journal in an appropriate area with a good chance of acceptance.

D.3 LENGTH AND FORMAT

The following length and style guidelines apply but your supervisor can advise on specific requirements:

Expected Length:	Standard format:	Typically between 40 and 50 pages of main text.
	Annotated Jupyter Notebook format:	Typically between 20 and 30 pages of main text with 10-15 pages of annotated Jupyter Notebooks.
	Scientific Paper:	Typically 10 pages.
Typing:	The report should be typed on A4 paper • using Times New Roman font, size 11 or 12 point, • 1.5 line spacing, • 2.5cm margins at top, bottom, left and right hand sides.	
Style:	The style of writing should be formal and impersonal, but attempting to interest a technically aware reader. Very detailed discussion of data-sets, simulations, software or intricate derivation of analytical results may be more appropriate in appendices.	
Page Numbering:	Page numbers must be included. The main body text should start at page 1.	

0.4 CITATIONS AND REFERENCES

References should acknowledge the contribution of others and indicate the level of awareness of students of work undertaken in their chosen field. Either the Harvard Referencing Style or the *IEEE citation style* for references should be used throughout the report. UL recommends using the software package EndNote for handling references. EndNote is a bibliographic management software program which allows you to create your own reference database and then to use that database to generate a bibliography of your papers, books and theses. It is available in Desktop and Web versions and a copy for installation can be obtained from the library.

0.5 SUBMITTING

Submitting the final project deliverables

Submission dates are hard and can be found in Table 1.

- Submit the electronic copy of the final dissertation in pdf format to your Brightspace project module MSc in AI Module - ED5005 named as follows, where **XY** are the initials of your supervisor and **xxxxxxxx** your id number:

  MScAI_Dissertation_**XY**_**xxxxxxxx**_FirstName_Surname.pdf

- Submit all auxiliary supporting material e.g. software, databases, data gathered etc. to Brightspace, compressed in a **zip** file named as follows (as agreed with supervisor):

  MScAI_Aux_**XY**_**xxxxxxxx**_FirstName_Surname.zip

- Please submit your dissertation abstract as a separate Word file for inclusion in a book of abstracts:

  MScAI_Abstract_**XY**_**xxxxxxxx**_FirstName_Surname.docx

0.6 SUPPORT

The Writing Centre at UL is a place for all students to get help on any aspect of their writing. There is a list of useful links and much more, including access to peer tutors.

APPENDIX E SUMMER YEAR 2: FINAL PROJECT PRESENTATION

A 15 minute demonstration/presentation/Q&A session (10min presentation + 5min discussion) will take place during the Summer semester emphasising the achievements and results. Students will present with the use of PowerPoint slides, unless agreed otherwise with supervisor beforehand. Supervisors can advise on the presentation format most suitable to individual projects.

Indicative dates can be found in Table 1. Students will be emailed with precise days and times in advance. Presentations will be online and webcams are a requirement.

APPENDIX F :GRADING

Masters projects will be graded based on the student's dissertation and final presentation using the below rubric. In recognition of the vast variety in dissertation projects and the need for flexibility in grading, supervisors and students can agree on different weightings of the rubric items during year 2. Weights can be changed by a maximum of 1 (on a scale of 10) and each item must have a minimum weight of 1. The sum of weights should remain 10.

	Weight
Final Report Structure and Appearance	1
Literature Survey/Background	1
Research Methodology and Approach	3
Research Findings and Conclusions	2
Quality of References	1
Presentation	2

APPENDIX G: MAJOR UPDATES

Since the last published version (February 2024) the following changes were made:

- Table 1 updated with dates for next academic year dates.
- Paragraph included in Introduction re showcasing knowledge and skills through code resulting from dissertation for professional progression.
- Addition of section 3.4 for Project Topic Change.
- Appendix E updated to clarify use of PowerPoint and webcams for presentation.